

Packet Tracer - Single-Area OSPFv2 Configuration (Instructor Version)

Instructor Note: Red font color or gray highlights indicate text that appears in the instructor copy only.

Addressing Table

Device	Interface	IP Address / Prefix
P2P-1	S0/1/0	10.0.0.1/30
	S0/1/1	10.0.0.9/30
	S0/2/0	10.0.0.13/30
P2P-2	S0/1/0	10.0.0.2/30
	S0/1/1	10.0.0.5/30
	G0/0/0	192.168.1.1/24
	G0/0/1	192.168.2.1/24
P2P-3	S0/1/0	10.0.0.6/30
	S0/1/1	10.0.0.10/30
	G0/0/0	192.168.3.1/28
BC-1	S0/1/0	10.0.0.14/30
	S0/1/1	64.0.100.2/30
	G0/0/0	10.0.1.1/29
BC-2	G0/0/0	192.168.4.1/30
	G0/0/1	10.0.1.2/29
BC-3	G0/0/0	192.168.5.1/24
	G0/0/1	10.0.1.3/29
Internet Server	NIC	203.0.113.100/24
PC 1	NIC	192.168.1.10/24
Laptop 1	NIC	192.168.2.20/24
Workgroup Server	NIC	192.168.3.14/28
PC 2	NIC	192.168.4.40/24

Device	Interface	IP Address / Prefix
PC 3	NIC	192.168.5.50/24

Objectives

Implement single-area OSPFv2 in both point-to-point and broadcast multiaccess networks.

Background

You are helping a network engineer test an OSPF set up by building the network in the lab where you work. You have interconnected the devices and configured the interfaces and have connectivity within the local LANs. Your job is to complete the OSPF configuration according to the requirements left by the engineer.

Use the information provided and the list of requirements to configure the test network. When the task has been successfully completed, all hosts should be able to ping the internet server.

Instructions

Configure the network to meet the requirements.

Requirements

Use process ID 10 for OSPF activation on all routers.

- Activate OSPF using network statements and inverse masks on the routers in the Headquarters network.
- Activate OSPF by configuring the interfaces of the network devices in the Data Service network, where required.
- Configure router IDs on the multiaccess network routers as follows:
 - BC-1: 6.6.6.6
 - BC-2: 5.5.5.5
 - BC-3: 4.4.4.4
- Configure OSPF so that routing updates are not sent into networks where they are not required.
- Configure router BC-1 with the highest OSPF interface priority so that it will always be the designated router of the multiaccess network.
- Configure a default route to the ISP cloud using the exit interface command argument.
- Automatically distribute the default route to all routers in the network.
- Configure the OSPF routers so that the Gigabit Ethernet interface cost will be 10 and the Fast Ethernet cost will be 100.
- Configure the OSPF cost value of P2P-1 interface Serial0/1/1 to 50.
- Configure the hello and dead timer values on the interfaces that connect P2P-1 and BC-1 to be twice the default values.

Answer Configurations

P2P-1

enable

configure terminal

```
interface Serial0/1/1
 ip ospf cost 50
interface Serial0/2/0
 ip ospf hello-interval 20
 ip ospf dead-interval 80
router ospf 10
 auto-cost reference-bandwidth 10000
 network 10.0.0.0 0.0.0.3 area 0
 network 10.0.0.8 0.0.0.3 area 0
 network 10.0.0.12 0.0.0.3 area 0
End
```

```
P2P-1>enable
P2P-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
P2P-1(config)#interface Serial0/1/1
P2P-1(config-if)#ip ospf cost 50
P2P-1(config-if)#interface Serial0/2/0
P2P-1(config-if)#ip ospf hello-interval 20
P2P-1(config-if)#ip ospf dead-interval 80
P2P-1(config-if)#router ospf 10
P2P-1(config-router)#auto-cost reference-bandwidth 10000
% OSPF: Reference bandwidth is changed.
      Please ensure reference bandwidth is consistent across all routers.
P2P-1(config-router)#
P2P-1(config-router)#network 10.0.0.0 0.0.0.3 area 0
P2P-1(config-router)#network 10.0.0.8 0.0.0.3 area 0
P2P-1(config-router)#network 10.0.0.12 0.0.0.3 area 0
P2P-1(config-router)#end
P2P-1#
%SYS-5-CONFIG_I: Configured from console by console
```

P2P-2

```
enable
configure terminal
router ospf 10
 passive-interface GigabitEthernet0/0/0
 passive-interface GigabitEthernet0/0/1
 auto-cost reference-bandwidth 10000
 network 10.0.0.0 0.0.0.3 area 0
 network 10.0.0.4 0.0.0.3 area 0
 network 192.168.1.0 0.0.0.255 area 0
 network 192.168.2.0 0.0.0.255 area 0
End
```

```
P2P-2>enable
P2P-2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
P2P-2(config)#passive-interface GigabitEthernet0/0/0
      ^
% Invalid input detected at '^' marker.

P2P-2(config)#router ospf 10
P2P-2(config-router)#passive-interface GigabitEthernet0/0/0
P2P-2(config-router)#passive-interface GigabitEthernet0/0/1
P2P-2(config-router)#auto-cost reference-bandwidth 10000
% OSPF: Reference bandwidth is changed.
    Please ensure reference bandwidth is consistent across all routers.
P2P-2(config-router)#network 10.0.0.0 0.0.0.3 area 0
P2P-2(config-router)#network 10.0.0.
12:12:51: %OSPF-5-ADJCHG: Process 10, Nbr 10.0.0.13 on Serial0/1/0 from LOADING
P2P-2(config-router)#network 10.0.0.4 0.0.0.3 area 0
P2P-2(config-router)#network 192.168.1.0 0.0.0.255 area 0
P2P-2(config-router)#network 192.168.2.0 0.0.0.255 area 0
P2P-2(config-router)#end
P2P-2#
%SYS-5-CONFIG_I: Configured from console by console
```

P2P-3

```
enable
configure terminal
router ospf 10
  passive-interface GigabitEthernet0/0/0
  auto-cost reference-bandwidth 10000
  network 10.0.0.4 0.0.0.3 area 0
  network 10.0.0.8 0.0.0.3 area 0
  network 192.168.3.0 0.0.0.15 area 0
End
```

```
P2P-3>enable
P2P-3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
P2P-3(config)#router ospf 10
P2P-3(config-router)#passive-interface GigabitEthernet0/0/0
P2P-3(config-router)#auto-cost reference-bandwidth 10000
% OSPF: Reference bandwidth is changed.
    Please ensure reference bandwidth is consistent across all routers.
P2P-3(config-router)#network 10.0.0.4 0.0.0.3 area 0
P2P-3(config-router)#netwo
12:18:24: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.2.1 on Serial0/1/0 from LOADING to FULL,
Loading
P2P-3(config-router)#network 10.0.0.8 0.0.0.3 area 0
P2P-3(config-router)#n
12:18:48: %OSPF-5-ADJCHG: Process 10, Nbr 10.0.0.13 on Serial0/1/1 from LOADING to FULL,
Loading Done
etwork
% Incomplete command.
P2P-3(config-router)#network 192.168.3.0 0.0.0.15 area 0
P2P-3(config-router)#end
P2P-3#
%SYS-5-CONFIG_I: Configured from console by console
```

BC-1

```
enable
configure terminal
interface GigabitEthernet0/0/0
 ip ospf priority 255
 ip ospf 10 area 0
interface Serial0/1/0
 ip ospf hello-interval 20
 ip ospf dead-interval 80
 ip ospf 10 area 0
router ospf 10
 router-id 6.6.6.6
 passive-interface Serial0/1/1
 auto-cost reference-bandwidth 10000
 default-information originate
 ip route 0.0.0.0 0.0.0.0 Serial0/1/1
End
```

```
BC-1>enable
BC-1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
BC-1(config)#interface GigabitEthernet0/0/0
      ^
% Invalid input detected at '^' marker.

BC-1(config)#interface GigabitEthernet0/0/0
BC-1(config-if)#ip ospf priority 255
BC-1(config-if)#ip ospf 10 area 0
BC-1(config-if)#interface Serial0/1/0
BC-1(config-if)#ip ospf hello-interval 20
BC-1(config-if)#ip ospf dead-interval 80
BC-1(config-if)#ip ospf 10 area 0
BC-1(config-if)#router
12:21:47: %OSPF-5-ADJCHG: Process 10, Nbr 10.0.0.13 on Serial0/1/0 from LOADING to FULL,
Loading Done

      ^
% Invalid input detected at '^' marker.

BC-1(config-if)#router ospf 10
BC-1(config-router)#router - id 6.6.6.6
      ^
% Invalid input detected at '^' marker.

BC-1(config-router)#router-id 6.6.6.6
BC-1(config-router)#passive-interface Serial0/1/1
BC-1(config-router)#auto-cost reference-bandwidth 10000
% OSPF: Reference bandwidth is changed.
Please ensure reference bandwidth is consistent across all routers.
BC-1(config-router)#default-information originate
BC-1(config-router)#ip route 0.0.0.0 0.0.0.0 Serial0/1/1
%Default route without gateway, if not a point-to-point interface, may impact performance
BC-1(config)#end
BC-1#
%SYS-5-CONFIG_I: Configured from console by console
|
```

BC-2

```
enable
configure terminal
interface GigabitEthernet0/0/0
 ip ospf 10 area 0
interface GigabitEthernet0/0/1
 ip ospf 10 area 0
router ospf 10
 router-id 5.5.5.5
 passive-interface GigabitEthernet0/0/0
 auto-cost reference-bandwidth 10000
End
```

```
BC-2>enable
BC-2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
BC-2(config)#interface GigabitEthernet0/0/0
BC-2(config-if)#ip ospf 10 area 0
BC-2(config-if)#interface GigabitEthernet0/0/1
BC-2(config-if)#ip ospf 10 area 0
BC-2(config-if)#router
12:29:21: %OSPF-5-ADJCHG: Process 10, Nbr 6.6.6.6 on GigabitEthernet0/0/1 from LOADING to FULL,
Loading Done
 ospf 10
BC-2(config-router)#router-id 5.5.5.5
BC-2(config-router)#passive-interface GigabitEthernet0/0/0
BC-2(config-router)#auto-cost reference-bandwidth 10000
% OSPF: Reference bandwidth is changed.
      Please ensure reference bandwidth is consistent across all routers.
BC-2(config-router)#end
BC-2#
%SYS-5-CONFIG_I: Configured from console by console
|
```

BC-3

```
enable
configure terminal
interface GigabitEthernet0/0/0
 ip ospf 10 area 0
interface GigabitEthernet0/0/1
 ip ospf 10 area 0
router ospf 10
 router-id 4.4.4.4
 passive-interface GigabitEthernet0/0/0
 auto-cost reference-bandwidth 10000
End
```

```
BC-3>enable
BC-3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
BC-3(config)#interface GigabitEthernet0/0/0
BC-3(config-if)#ip ospf 10 area 0
BC-3(config-if)#interface GigabitEthernet0/0/1
BC-3(config-if)#ip ospf 10 area 0
BC-3(config-if)#router-id 4.4.4
12:32:20: %OSPF-5-ADJCHG: Process 10, Nbr 5.5.5.5 on GigabitEthernet0/0/1 from LOADING to FULL,
Loading Done
.
12:32:21: %OSPF-5-ADJCHG: Process 10, Nbr 6.6.6.6 on GigabitEthernet0/0/1 from LOADING to FULL,
Loading Done
4
      ^
% Invalid input detected at '^' marker.

BC-3(config-if)#router-id 4.4.4.4
      ^
% Invalid input detected at '^' marker.

BC-3(config-if)#
BC-3(config-if)#router ospf 10
BC-3(config-router)#router-id 4.4.4.4
BC-3(config-router)#passive-interface GigabitEthernet0/0/0
BC-3(config-router)#auto-cost reference-bandwidth 10000
% OSPF: Reference bandwidth is changed.
      Please ensure reference bandwidth is consistent across all routers.
BC-3(config-router)#end
BC-3#
%SYS-5-CONFIG_I: Configured from console by console
|
```